

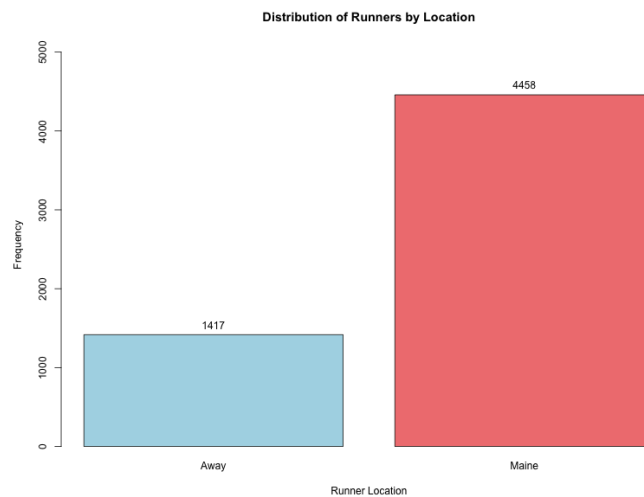
CS6313- Mini Project 2

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Section 1

1. (12 points) Consider the dataset roadrace.csv posted on eLearning. It contains observations on 5875 runners who finished the 2010 Beach to Beacon 10K Road Race in Cape Elizabeth, Maine. You can read the dataset in R using read.csv function.

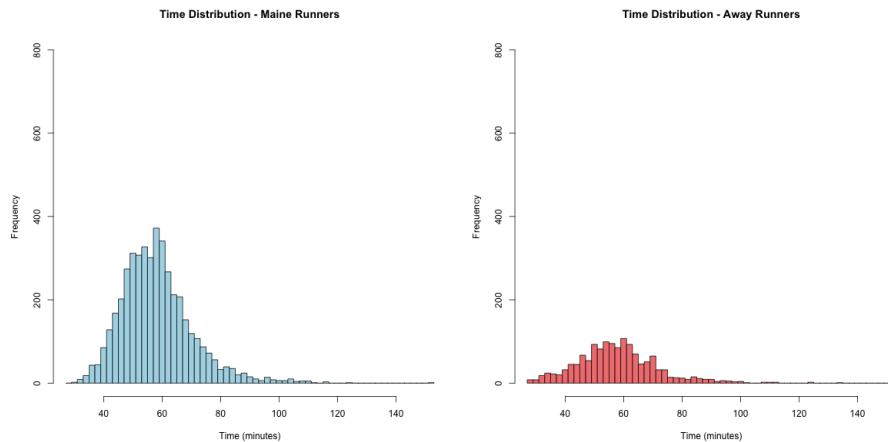
(a) Create a bar graph of the variable Maine, which identifies whether a runner is from Maine or from somewhere else (stated using Maine and Away). You can use barplot function for this. What can we conclude from the plot? Back up your conclusions with relevant summary statistics.



The bar graph reveals that the 2010 Beach to Beacon 10K Road Race had significantly more participants from Maine than from other locations. Out of 5,875 total runners, 4,458 were Maine residents while only 1,417 were from away. This indicates that the race attracts a predominantly local audience, with Maine runners comprising approximately 75.9% of all participants compared to just 24.1% from outside Maine.

The substantial difference between the two groups (more than three times as many Maine runners) suggests that this race serves primarily as a community event for local runners, though it does attract a notable number of out-of-state participants as well.

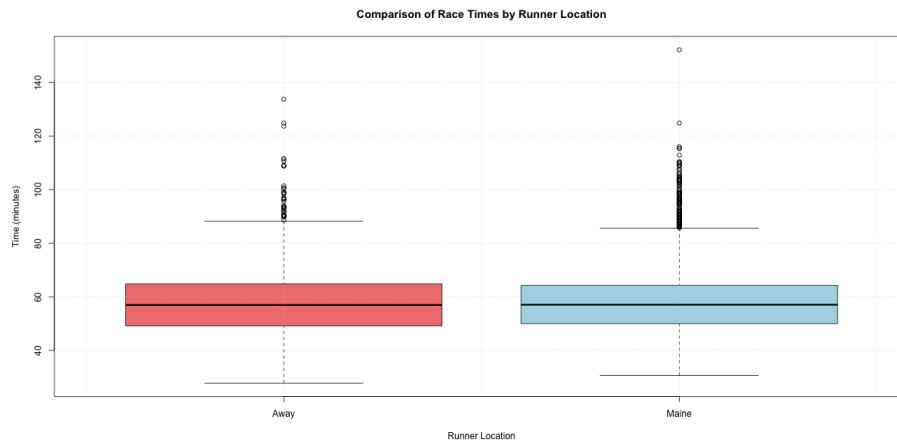
b) Create two histograms the runners' times (given in minutes) — one for the Maine group and the second for the Away group. Make sure that the histograms on the same scale. What can we conclude about the two distributions? Back up your conclusions with relevant summary statistics, including mean, standard deviation, range, median, and interquartile range.



Both distributions are approximately right-skewed (positively skewed), with a longer tail extending toward higher completion times. However, both distributions are roughly unimodal with a single peak, suggesting most runners cluster around similar finishing times.

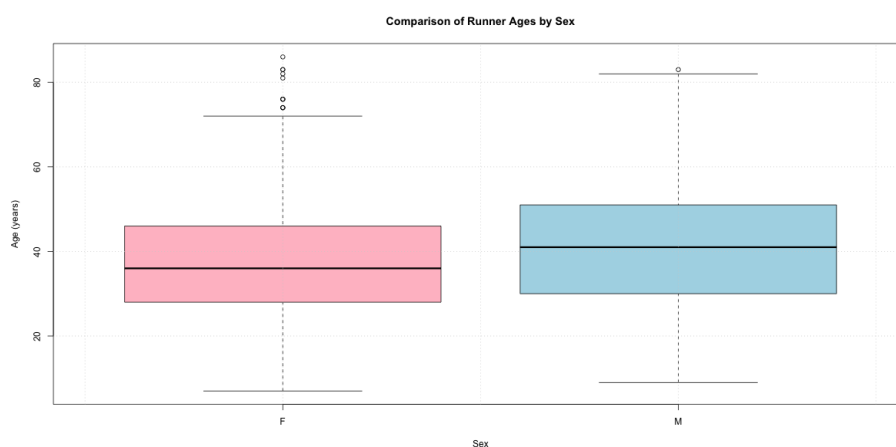
The two groups have remarkably similar central tendencies. Maine runners have a mean time of 58.20 minutes and median of 57.03 minutes. Away runners have a mean time of 57.82 minutes and median of 56.92 minutes.

1c) Repeat (b) but with side-by-side boxplots.



The boxplots reinforce the conclusions we came to from the histograms. Both sets of runners have almost identical interquartile ranges. The outliers are also in similar ranges with the only major difference being that there are significantly more outliers from Maine. This difference can be simply explained by the difference in number of runners from Maine versus from out of state.

1d) Create side-by-side boxplots for the runners' ages (given in years) for male and female runners. What can we conclude about the two distributions? Back up your conclusions with relevant summary statistics, including mean, standard deviation, range, median, and interquartile range.

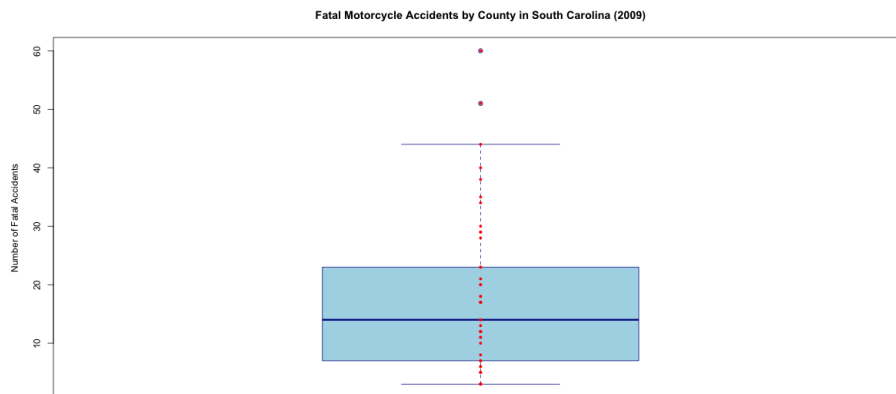


Male runners have a higher median age (41 years) compared to female runners (36 years). The mean ages show a similar pattern: males average 40.45 years while females average 37.24 years. This 5-year difference indicates that male participants in this race tend to be older than female participants.

Male runners show a wider interquartile range ($IQR = 21$ years) compared to females ($IQR = 18$ years). Standard deviation values also support this: males show 14.0 years SD vs. females' 12.3 years SD. Male runners display more age diversity in their central distribution.

Female distribution appears more symmetric, with the median (36) roughly centered in the box. Male distribution shows slight right skewness, with more spread in the upper half of the box. Both groups show some concentration in the 30-50 age range.

2. (8 points) Consider the dataset `motorcycle.csv` posted on eLearning. It contains the number of fatal motorcycle accidents that occurred in each county of South Carolina during 2009. Create a boxplot of data and provide relevant summary statistics. Discuss the features of the data distribution. Identify which counties may be considered outliers. Why might these counties have the highest numbers of motorcycle fatalities in South Carolina?



Horry County contains Myrtle Beach, a nationally popular spring-break and bike-rally destination. Large seasonal influx of motor-cyclists on unfamiliar roads, beach-town traffic, higher speeds on US-17 and SC-31 elevate risk.

Greenville County is the most populous county in the Up-state and includes much of the I-85 corridor. More registered motorcycles, higher vehicle-miles-travelled, plus urban interstate exchanges are known high-risk settings for fatal bike crashes. Both counties also have major medical centres, so crashes occurring within the county are more likely to be recorded there even if riders live elsewhere, reinforcing their top positions.

Overall, The 2009 data are strongly right-skewed with a low of 3 and two clear high outliers—Horry and Greenville—whose tourism volume and urban interstate traffic explain their disproportionate numbers of motorcycle fatalities.

Section 2

Code and output are provided in section 2.

```

# Reset variables
rm(list = ls())

# Load required libraries
library(ggplot2)

# Read the dataset
roadrace <- read.csv("roadrace.csv")

# =====
# QUESTION 1(a): Bar graph of Maine variable
# =====

# Create frequency table for Maine variable
maine_table <- table(roadrace$Maine)
print("Frequency table for Maine variable:")
print(maine_table)

# Calculate proportions
maine_prop <- prop.table(maine_table)
print("Proportions:")
print(maine_prop)

png("barplot_maine.png", width = 800, height = 600) # Open file for saving

# Create bar graph
bp <- barplot(maine_table,
  main = "Distribution of Runners by Location",
  xlab = "Runner Location",
  ylab = "Frequency",
  col = c("lightblue", "lightcoral"),
  ylim = c(0, max(maine_table) * 1.15)
)

# Add count labels on top of bars
text(
  x = bp,
  y = maine_table,
  labels = maine_table,
  pos = 3
)

dev.off() # Close and save the file

```

```

# =====
# QUESTION 1(b): Histograms of runners' times by location
# =====

# Subset data by Maine status
maine_runners <- roadrace[roadrace$Maine == "Maine", ]
away_runners <- roadrace[roadrace$Maine == "Away", ]

# Summary statistics for Maine runners
cat("\n=== Summary Statistics for Maine Runners (Time in minutes) ===\n")
cat("Mean:", mean(maine_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Standard Deviation:", sd(maine_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Range:", range(maine_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Median:", median(maine_runners$Time..minutes., na.rm = TRUE), "\n")
cat("IQR:", IQR(maine_runners$Time..minutes., na.rm = TRUE), "\n")

# Summary statistics for Away runners
cat("\n=== Summary Statistics for Away Runners (Time in minutes) ===\n")
cat("Mean:", mean(away_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Standard Deviation:", sd(away_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Range:", range(away_runners$Time..minutes., na.rm = TRUE), "\n")
cat("Median:", median(away_runners$Time..minutes., na.rm = TRUE), "\n")
cat("IQR:", IQR(away_runners$Time..minutes., na.rm = TRUE), "\n")

# Determine common scale for histograms
time_range <- range(roadrace$Time..minutes., na.rm = TRUE)
breaks_seq <- seq(
  ~~~~~
  from = floor(time_range[1]),
  to = ceiling(time_range[2]),
  by = 2
)

png("barplot_timeDistribution.png", width = 1200, height = 600)

# Create histograms side by side
par(mfrow = c(1, 2))

# Histogram for Maine runners
hist(maine_runners$Time..minutes.,
  ~~~~~
  breaks = breaks_seq,
  main = "Time Distribution - Maine Runners",
  xlab = "Time (minutes)",
  ylab = "Frequency",
  col = "#lightblue",
  xlim = time_range,
  ylim = c(0, 800)
)

```

```
)  
dev.off() # Close and save the file  
  
# =====  
# QUESTION 1(c): Side-by-side boxplots for time by location  
# =====  
  
png("boxplot_timeDistribution.png", width = 1200, height = 600)  
boxplot(Time..minutes. ~ Maine,  
  data = roadrace,  
  main = "Comparison of Race Times by Runner Location",  
  xlab = "Runner Location",  
  ylab = "Time (minutes)",  
  col = c("lightcoral", "lightblue"),  
  horizontal = FALSE  
)  
  
# Add grid for easier reading  
grid()  
  
dev.off() # Close and save the file
```



```

# =====
# QUESTION 1(d): Side-by-side boxplots for age by sex
# =====

# Summary statistics for Male runners
male_runners <- roadrace[roadrace$Sex == "M", ]
female_runners <- roadrace[roadrace$Sex == "F", ]

cat("\n=== Summary Statistics for Male Runners (Age in years) ===\n")
cat("Mean:", mean(male_runners$Age, na.rm = TRUE), "\n")
cat("Standard Deviation:", sd(male_runners$Age, na.rm = TRUE), "\n")
cat("Range:", range(male_runners$Age, na.rm = TRUE), "\n")
cat("Median:", median(male_runners$Age, na.rm = TRUE), "\n")
cat("IQR:", IQR(male_runners$Age, na.rm = TRUE), "\n")

cat("\n=== Summary Statistics for Female Runners (Age in years) ===\n")
cat("Mean:", mean(female_runners$Age, na.rm = TRUE), "\n")
cat("Standard Deviation:", sd(female_runners$Age, na.rm = TRUE), "\n")
cat("Range:", range(female_runners$Age, na.rm = TRUE), "\n")
cat("Median:", median(female_runners$Age, na.rm = TRUE), "\n")
cat("IQR:", IQR(female_runners$Age, na.rm = TRUE), "\n")

png("boxplot_comparison_runner_age_by_sex.png", width = 1200, height = 600)
# Create side-by-side boxplots
boxplot(Age ~ Sex,
  data = roadrace,
  main = "Comparison of Runner Ages by Sex",
  xlab = "Sex",
  ylab = "Age (years)",
  col = c("pink", "lightblue"),
  horizontal = FALSE
)

# Add grid for easier reading
grid()

dev.off() # Close and save the file

```

```

# Question 2
# Analysis of Fatal Motorcycle Accidents in South Carolina (2009)

# Read the dataset
motorcycle <- read.csv("motorcycle.csv")

# View the first few rows
head(motorcycle)

# Remove counties with 0 accidents (OTHER and UNKNOWN)
motorcycle_filtered <- motorcycle[motorcycle$Fatal.Motorcycle.Accidents > 0, ]

# =====
# Create boxplot
# =====
png("boxplot_Fatal_Motorcycle_Accidents.png", width = 1200, height = 600)

boxplot(motorcycle_filtered$Fatal.Motorcycle.Accidents,
        main = "Fatal Motorcycle Accidents by County in South Carolina (2009)",
        ylab = "Number of Fatal Accidents",
        col = "#lightblue",
        border = "#darkblue")

# Add points to show individual counties
points(rep(1, nrow(motorcycle_filtered)),
       motorcycle_filtered$Fatal.Motorcycle.Accidents,
       pch = 16, col = "red", cex = 0.7)

dev.off() # Close and save the file

# =====
# Summary Statistics
# =====
cat("\n==== SUMMARY STATISTICS =====\n")
cat("Mean:", mean(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("Median:", median(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("Standard Deviation:", sd(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("Range:", range(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("IQR (Interquartile Range):", IQR(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("Q1 (25th percentile):", quantile(motorcycle_filtered$Fatal.Motorcycle.Accidents, 0.25), "\n")
cat("Q3 (75th percentile):", quantile(motorcycle_filtered$Fatal.Motorcycle.Accidents, 0.75), "\n")
cat("Min:", min(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")
cat("Max:", max(motorcycle_filtered$Fatal.Motorcycle.Accidents), "\n")

```

```

# Five number summary
cat("\nFive Number Summary:\n")
print(fivenum(motorcycle_filtered$Fatal.Motorcycle.Accidents))

# =====
# Identify Outliers
# =====
Q1 <- quantile(motorcycle_filtered$Fatal.Motorcycle.Accidents, 0.25)
Q3 <- quantile(motorcycle_filtered$Fatal.Motorcycle.Accidents, 0.75)
IQR_value <- IQR(motorcycle_filtered$Fatal.Motorcycle.Accidents)

# Calculate outlier boundaries
lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

cat("\n==== OUTLIER ANALYSIS =====\n")
cat("Lower bound for outliers:", lower_bound, "\n")
cat("Upper bound for outliers:", upper_bound, "\n")

# Identify outliers
outliers <- motorcycle_filtered[motorcycle_filtered$Fatal.Motorcycle.Accidents > upper_bound |
                                motorcycle_filtered$Fatal.Motorcycle.Accidents < lower_bound, ]

cat("\nCounties identified as outliers:\n")
print(outliers)

# Sort to see highest values
cat("\nTop 5 counties with most fatal accidents:\n")
top_counties <- motorcycle_filtered[order(-motorcycle_filtered$Fatal.Motorcycle.Accidents), ][1:5, ]
print(top_counties)

```